VivaGuard

Blood Pressure Monitor
User Manual

Model: AOJ-35C

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INTRODUCTION

Thank you for purchasing Portable Blood Pressure Monitor. The monitor is convenient to use at home, at work and during vacation, perfect for people who need daily blood pressure measurement.
The Monitor uses the oscillometric method of blood pressure measurement. It is intended for use by medical professionals or at home to monitor and display diastolic, systolic blood pressure and pulse rate, With an air wrist cuff following instructions of "HOW TO APPLY THE WRIST CUFF".
The expected lifespan of the device is 5 years.
The device complies with the electromagnetic compatibility requirement of EN60601-1-2 and safety standards of EN60601-1 and performance of IEC 80601-2-30 as specified in EEC directive 93/42/EEC.

NOTES ON SAFETY

* The warning signs and sample icons shown here are listed for your safe and correct use of the unit, so as to prevent injuries or damages to the device.
* The icons and meanings are as follows:

Examples of signs

The ⃠ icon indicates prohibitions (what you should not do). Matters involving actual prohibitions are indicated by text or pictures in or near ⃠. The left icon refers to "general prohibition".

The ❶ icon indicates something that is compulsory (what must always be observed).
Matters involving actual compulsory actions are indicated by text or pictures in or near ❶. The left icon refers to "general compulsion".

The ⃠ icon indicates something can't be disassembled or "Don't disassemble". Matters involving actual compulsory actions are indicated by text or pictures in or near ⃠. The left icon refers to "general prohibition".

Type BF Applied part

CE 0123 CE mark

Please refer to the instructions for use

RoHs RoHs mark

Indicates a medical device that needs to be protected from moisture.

Vertical upward

The equipment should be sent to the special agenciesaccording to local regulation for separate collection afterits useful life.

Keep out of the sun

IP22 The device is protected against splashing water. Water splashed against the enclosure from any direction shall have no harmful effects.

Date of manufacture

Manufacturer

SN Serial number

LOT Lot number

EC REP EU authorized representative

⚠ Caution

Patient must follow doctor's instruction and should not perform self-judgment and self-treatment up to the measured result, Self-diagnosis of measured results and treatment are dangerous, The device should not be used to judge illness, first aid or continuously monitor measuring.
This device can not be used for Patient and surgical care. It can be used in household or fixed places only.The patient is an intended operator, and all the functions can be safely used
Please press "on/off " button to stop work when you feel uncomfortable with the wrist, or if the air is inflating abnormally without stop.

Do not let a child below 12 years old or the people who can't express intention to use the device. it should be accompanied by the Adult to avoid acciden or trouble.
Do not use the unit for purpose other than measuring blood pressure to avoid accident or trouble.
Please do not use mobile phone around the device. Please do not use the device around the magnetic field.
The device is prohibited from being used during movement.

Do not use the device in the Magnetic Resonance (MR) environment.
Do not use the equipment in outdoor or shower rooms.
Don't near active HF surgical equipment and the RF shielded room of an ME system for magnetic resonance imaging, where the intensity of EM disturbances is high.
Use of this equipment adjacent to or stacked with other equipment should be avoided because it could result in improper operation. If such use is necessary, this equipment and the other equipment should be observed to verify that they are operating normally.
Use of accessories, transducers and cables other than those specified or provided by the manufacturer of this equipment could result in increased electromagnetic emissions or decreased electromagnetic immunity of this equipment and result in improper operation."
Portable RF communications equipment (including peripherals such as antenna cables and external antennas) should be used no closer than 30 cm (12 inches) to any part of the equipment, including cables specified by the manufacturer. Otherwise, degradation of the performance of this equipment could result.
Do not disassemble, repair, or remodel the main unit or the wrist cuff of the blood pressure monitor, to avoid incorrect performance of unit.

Requests from Manufacturer

Make sure there is no connection tubing kinking before start measuring to avoid any injury to patient.

For any patient, do not measure more than 3 times continuously, it should be at least above 5 minutes of interval rest between any two measurements, otherwise will cause extravasated blood.
Do not measure your blood pressure over 6 times each day.
Do not apply the cuff over a wound as this can cause further injury.
Do not measure on the wrist which is on the side of a mastectomy, otherwise it could cause injury.
Observe the air pressure value from the VA display
When measuring, it could not exceed 295 mmHg, otherwise Please press "START/STOP" button to stop.
Do not use force to bend the wrist cuff or the air tube.
Do not knock or drop the main unit.
Always use the specified accessories in the manual, the use of other parts not approved by the manufacturer may cause faults or injuries.
For service information, parts list etc., please contact the dealer.
Dispose of the monitor when its service life is reached. Follow local regulations regarding the disposal of such product.
When the performance changes (such as: inaccurate measurement or abnormal display), please stop using it immediately and contact the after-sales service personnel in time.

Do not leave the small parts where children can reach them. Children may swallow them. If a child accidentally swallows them, please contact a doctor immediately.
Not servicing and maintenance while the ME EQUIPMENT is in use.
The user can maintain the product, the maintenance method is described in the maintenance instructions of manual.
Stop using the equipment immediately, if it is in contact with water.

INTENDED USE / INSTRUCTIONS FOR USE

The Wrist Blood Pressure Monitor is intended to measure the systolic pressure and diastolic pressure, as well as the pulse rate of adult person via non-invasive oscillometric technique by an inflatable cuff wrapped around the wrist at medical facilities or at home.

CONTRAINDICATION

No known contraindication existing.

ABOUT BLOOD PRESSURE

1. What is blood pressure?

Blood pressure is the force exerted by blood against the walls of the arteries. Systolic

pressure occurs when the heart contracts. Diastolic pressure occurs when the heart expands.

Blood pressure is measured in millimeters of mercury (mmHg). One's natural blood pressure is represented by the fundamental pressure, which is measured first thing in the morning while one is still at rest and before eating.

2. What is hypertension and how is it controlled?

Hypertension, an abnormally high arterial blood pressure, if left unattended, can cause many health problems including stroke and heart attack.

Hypertension can be controlled by altering lifestyle, avoiding stress and with medication under a doctor's supervision.

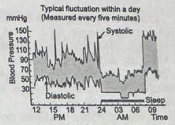
To prevent hypertension or keep it under control:

- Do not smoke
- Exercise regularly
- Maintain proper weight
- Reduce salt and fat intake
- Have regular physical checkups

3. Why measure blood pressure at home?

Blood pressure measured at a clinic or doctor's office may cause apprehension and produce an elevated reading, 25 to 30 mmHg higher than that measured at home, Home measurement reduces the effects of outside influences on blood pressure readings, supplements the doctor's readings and provides a more accurate, complete blood pressure history.

4. Blood pressure variations



An individual's blood pressure varies greatly on a daily and seasonal basis. It may vary by 30 to 50 mmHg due to various conditions during the day. In hypertensive individuals, variations are even more pronounced.

Normally, the blood pressure rises while at work or play and falls to its lowest levels during sleep. so, do not be overly concerned by the results of one measurement.

Take measurements at the same time everyday using the procedure described in this manual,and know your normal blood pressure.Many readings give a more comprehensive blood pressure history.

Be sure to note date and time when recording your blood pressure. Consult your doctor to interpret your blood pressure data.

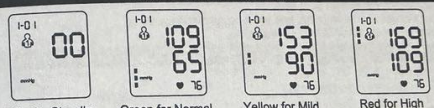
PRECAUTIONS BEFORE USE

1. If you are taking medication, consulting with your doctor to determine the most appropriate time to measure your blood pressure. NEVER change a prescribed medication without first consulting with your doctor.

2. For people with irregular or unstable peripheral circulation problems due to diabetes, liver disease, hardening of the arteries, etc., there may be fluctuation inblood pressure values measured at the upper arm versus at the wrist.
3. Measurements may be impaired if this device is used near televisions, microwave ovens, X-ray, mobile phone equipment or other devices with strong electrical fields.To prevent such interference, use the monitor at a sufficient distance from such devices or turn them off.
4. Before using, should wash your hands.
5. Do not measure on the arm which simultaneously used monitoring ME Equipment,otherwise it could cause loss of function.
6. Consult your doctor if the unexpected readings are obtained, also please refer to "Trouble shooting" of the manual.
7. Portable RF communications equipment (including peripherals such as antenna cables and external antennas) should be used no closer than 30 cm (12 inches) to any part of the this device, including cables specified by the manufacturer. Otherwise, degradation of the performance of this equipment could result.
8. When ambient temperature is less than 5℃ or higher than 40℃, The time from minimum storage temperature to use is at least 1 hour; The time from maximum storage temperature is at least 2 hour.

9.The reading is probably a little lower than measured in the hospital due to the steady mood at home.

FOUR COLORS READINGS



White for Standby | Green for Normal | Yellow for Mild High Pressure or Hypotension | Red for High Pressure

Systolic Pressure (mmHg)	Diastolic Pressure (mmHg)	Color of Reading	Hierarchical Relationship
≥160	≥100	red	and (or)
140-159	90-99	yellow	and (or)
90-139	60-89	green	and (or)
< 90	< 60	yellow	and

⚠ Note : Consult your doctor if unexpected readings are obtained.

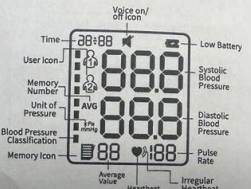
FEATURES OF THE PRODUCT

1. 2 users, 99 sets of memories for each user.
2. Large HD LCD touch screen.
3. 4-Color Readings with Backlight
4. Easy to use, visual warning, and voice broadcasting optional.
5. Auto power off within 20s to save power.

PARTS IDENTIFICATION

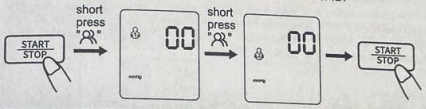
Composition: Main unit, cuff.

① HD LCD Touch Screen
② Previous/Page-up Touch button
③ Wrist Cuff
④ Next/Page-down Touch button
⑤ Memory Button
⑥ START/STOP Button
⑦ User Selection/General Setting Button



USER MODE

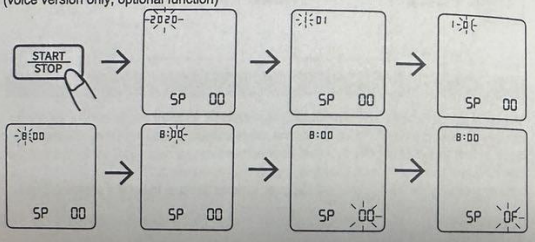
In standby mode, short press the "👤" button to enter the User ID selection interface. Then press the "👤" button to switch and select User ID.



DATE TIME AND MUTE FUNCTION SETUP

The Wrist Blood Pressure Monitor automatically stores up to 99 measurement values for each user. The memory function stores the measurement values with the date and time. Set the monitor to the current date and time before taking a measurement for the first time. The date and time can not be changed during a measurement.
1.Keep pressing the "👤" key until the year number flashes, to enter the setting mode.

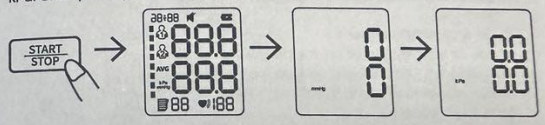
2. Press "Previous/Next Touch Button" to adjust the year, then press" 👤 " again to save your setting and enter the month setting mode.
3. Press "Previous/Next Touch Button" to adjust the month. Follow the same steps to adjust date/hour/minute.
4. After the time setting is finished, it will automatically enter the mute function setting (the screen will display "SP"). Short press Previous/Next Touch Button to switch ON/OFF, "ON" means voice on, "OFF" means mute mode on, short press " 👤 " button to confirm. (voice version only, optional function)



BLOOD PRESSURE UNIT CONVERSION

There are two kinds of units of blood pressure display, mmHg (mmHg) and kPa (kPa). mmHg (mmHg) is set as default.

In standby mode, keep pressing "START/STOP" button for around 5 seconds to enter the unit selection, and press "Previous/Next Touch Button" to switch mmHg/kPa. Short press "👤" to confirm the selection.



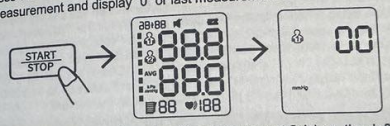
HOW TO APPLY THE WRIST CUFF

1. Fasten the wrist cuff
   1) Place the wrist cuff on left bare wrist by leaving clearance of 0.4-0.6 inch between the edge of the wrist cuff and the bottom of the palm as shown in the figure.
   2) Fasten the wrist cuff tightly by using the Velcro Strip.
2. Tips for proper measurements
   - Relax for 5-10 minutes before measurement.
   - Sit comfortably and rest your wrist on the table.
   - Raise your hand to help wrist cuff reach same level as your heart.
   - Keep still and quiet during measurement.
   - Do not measure after physical exercise or a bath.
   - Take Measurement at the same time every day.

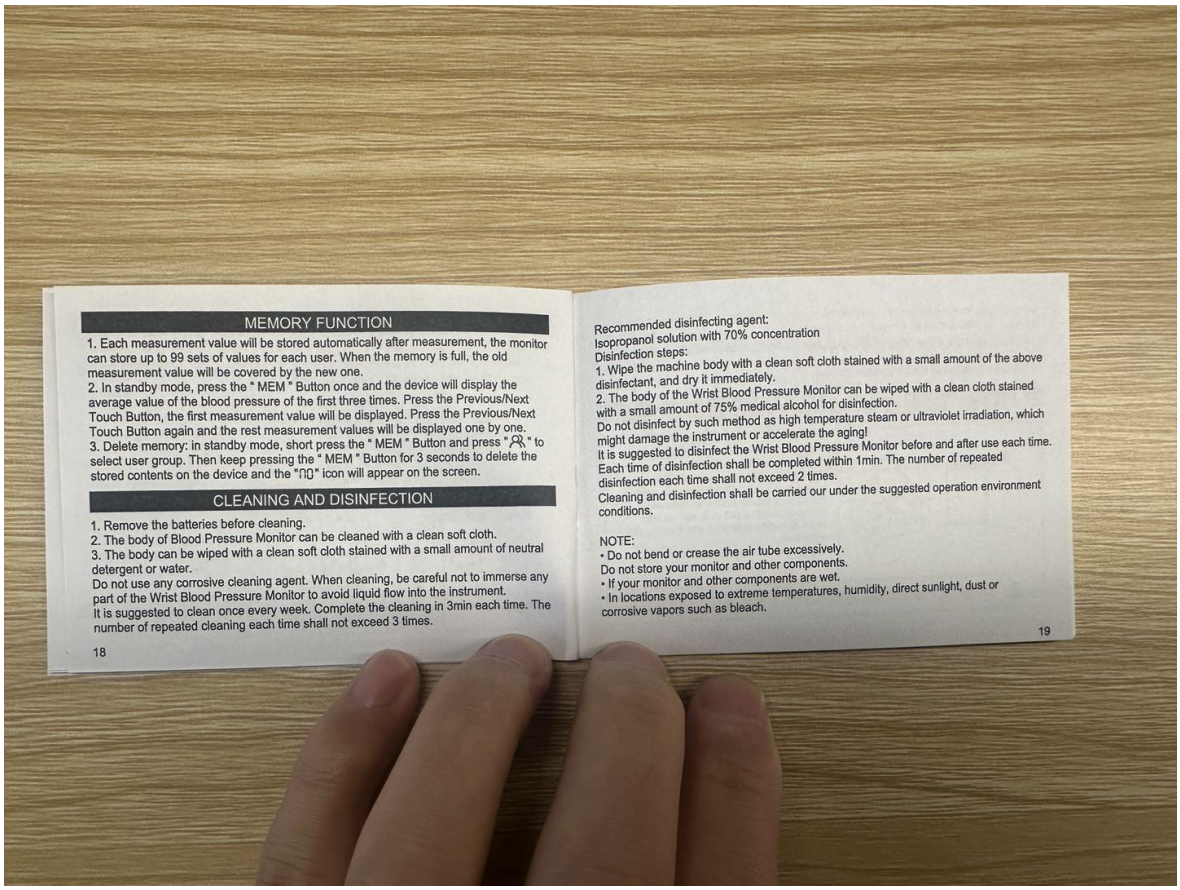
⚠ Please use the original cuff that has passed the clinical trial! Any replacement of original component with the component not provided by the manufacturer might result in measurement error.
When measuring, please keep your sitting posture upright and not cross your legs.

HOW TO TAKE A MEASUREMENT

1. Fasten the wrist cuff following the instruction of "HOW TO APPLY THE WRIST CUFF."
2. Press the "START/STOP" button. After all icons are turned on, monitor will start inflating for measurement and display "0" or last measurement record.



3. Check the measured values after the measurement finishes, the deflation mark flashes and the wrist cuff deflates.



MEMORY FUNCTION

1. Each measurement value will be stored automatically after measurement, the monitor can store up to 99 sets of values for each user. When the memory is full, the old measurement value will be covered by the new one.
2. In standby mode, press the " MEM " Button once and the device will display the average value of the blood pressure of the first three times. Press the Previous/Next Touch Button, the first measurement value will be displayed. Press the Previous/Next Touch Button again and the rest measurement values will be displayed one by one.
3. Delete memory: in standby mode, short press the " MEM " Button and press "👤" to select user group. Then keep pressing the " MEM " Button for 3 seconds to delete the stored contents on the device and the "∩" icon will appear on the screen.

CLEANING AND DISINFECTION

1. Remove the batteries before cleaning.
2. The body of Blood Pressure Monitor can be cleaned with a clean soft cloth.
3. The body can be wiped with a clean soft cloth stained with a small amount of neutral detergent or water.

Do not use any corrosive cleaning agent. When cleaning, be careful not to immerse any part of the Wrist Blood Pressure Monitor to avoid liquid flow into the instrument.
It is suggested to clean once every week. Complete the cleaning in 3min each time. The number of repeated cleaning each time shall not exceed 3 times.

Recommended disinfecting agent:
Isopropanol solution with 70% concentration
Disinfection steps:
1. Wipe the machine body with a clean soft cloth stained with a small amount of the above disinfectant, and dry it immediately.
2. The body of the Wrist Blood Pressure Monitor can be wiped with a clean cloth stained with a small amount of 75% medical alcohol for disinfection.

Do not disinfect by such method as high temperature steam or ultraviolet irradiation, which might damage the instrument or accelerate the aging!
It is suggested to disinfect the Wrist Blood Pressure Monitor before and after use each time.
Each time of disinfection shall be completed within 1min. The number of repeated disinfection each time shall not exceed 2 times.
Cleaning and disinfection shall be carried our under the suggested operation environment conditions.

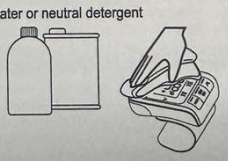
NOTE:
- Do not bend or crease the air tube excessively.

Do not store your monitor and other components.
- If your monitor and other components are wet.
- In locations exposed to extreme temperatures, humidity, direct sunlight, dust or corrosive vapors such as bleach.

- In locations exposed to vibrations or shocks.
- If your monitor and other components are wet.
- In locations exposed to extreme temperatures, humidity, direct sunlight, dust or corrosive vapors such as bleach.
- In locations exposed to vibrations or shocks.

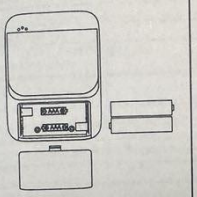
UPKEEP AND MAINTENANCE

water or neutral detergent



- Always keep the surface of Blood Pressure Monitor clean and tidy, helpful to prolong the service life of Blood Pressure Monitor.
- If the host is dirty, please wipe with a dry soft cloth. If the dirt cannot be eliminated easily, wipe with a soft cloth stained with water or neutral detergent, and then dry with a dry cloth.
-No maintenance or maintenance required when using the device.

⚠ Do not allow water or other liquids to flow into the host.



Install the battery
- Open the battery cover according to the method shown in the figure.
- Place 2 AAA dry batteries in the battery compartment, and pay attention to the electrode indication of the batteries. Install the battery as indicated in the picture right under this sentence.

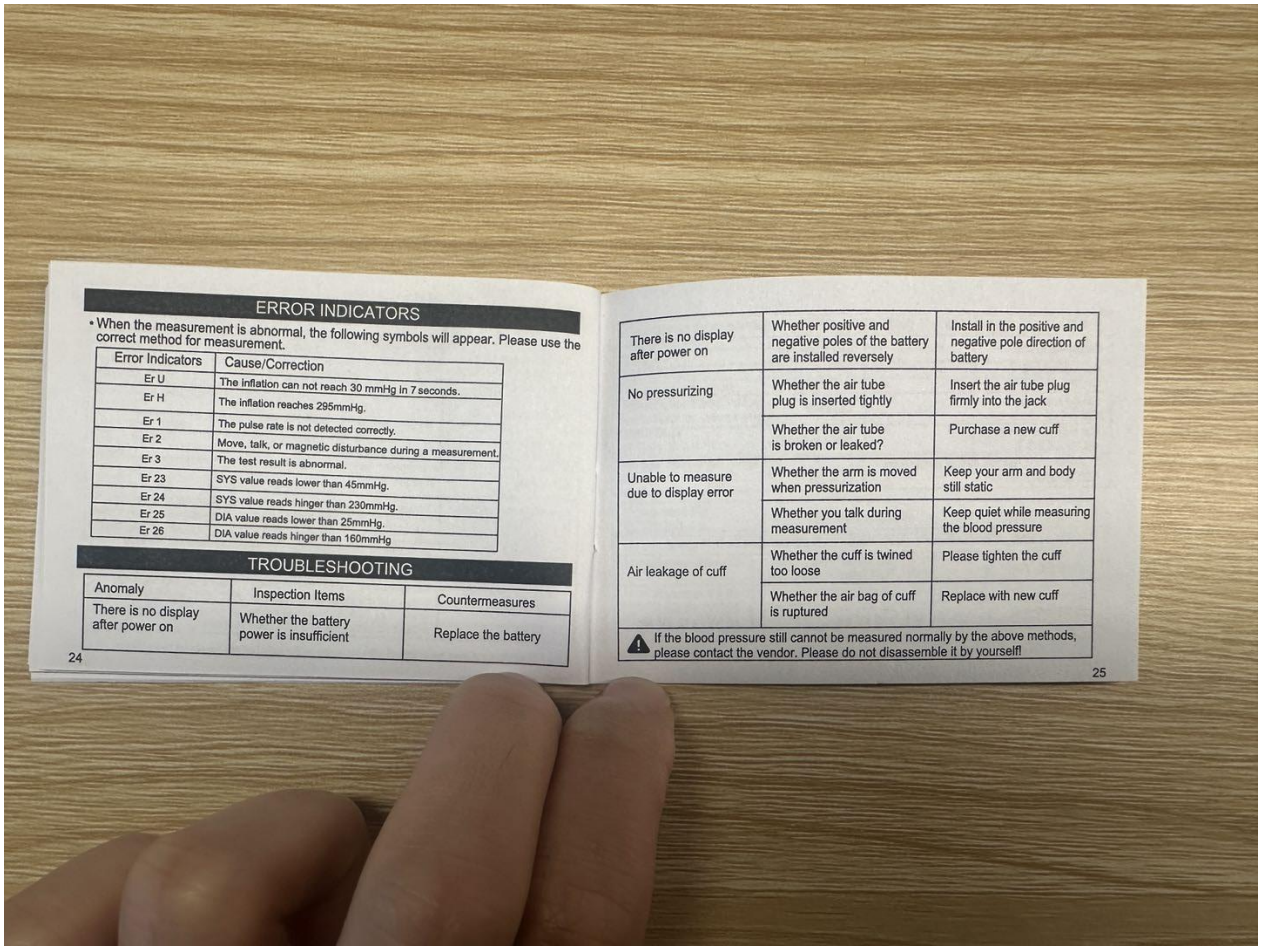
Battery replacement
- If the product is not used for a long time (over 3 months), please take out the batteries.

SPECIFICATIONS

Measuring Method	Oscillometric Method
Display	Large LCD Screen
Pneumatic pressure measuring range	0~295 mmHg (0~39.3Kpa)
Overpressure protection limit Adult	300 mmHg
Measurement range: Blood Pressure	SYS: 45-230 mmHg; DIA: 25-160 mmHg; Pulse rate: 40-199 bpm
Accuracy	Blood Pressure: ±3 mmHg(±0.4 Kpa) ; Pulse Rate: ± 5 bpm
Memory	2*99 Memories
Power Source	2*AAA batteries
Outer Dimensions	Approximately 3.13" (w) x 2.75" (h) x 1.05" (d) (79.6mm x 70mm x 26.8mm)
Screen Size	46x42mm (2.5 inches)
Console Weight	Approximately 4.5 oz. (126g) without batteries
Anti Electronic Shock Degree	Type BF

Anti Electronic Shock Type	Internally Power Supply	
Protection against harmful ingress of water or particular matter	IP22	
Wrist Circumference Range	Approximately 5.31"~7.68" (135mm~195mm)	
Operation mode	Continuous operation	
Service life	5 years	
Operating environment	Temperature Condition: 5 ℃-40 ℃	If stored or used beyond the designated temperature and humidity range, it will not be used properly
	Humidity Condition: 15%-90%RH	
	Atmospheric Condition: 70kPa-106kPa	
Transportation and Storage Environment	Avoid strong impact, direct impact, exposure or rain during transportation. The packaged Blood Pressure Monitor shall be stored indoors at the temperature of -20℃~55℃ and the relative humidity of 10%~93% ,atmospheric Condition: 70kPa-106kPa. without corrosive gas and with good ventilation.	

* Specifications may be changed without notice in the event of improvement being made.



ERROR INDICATORS

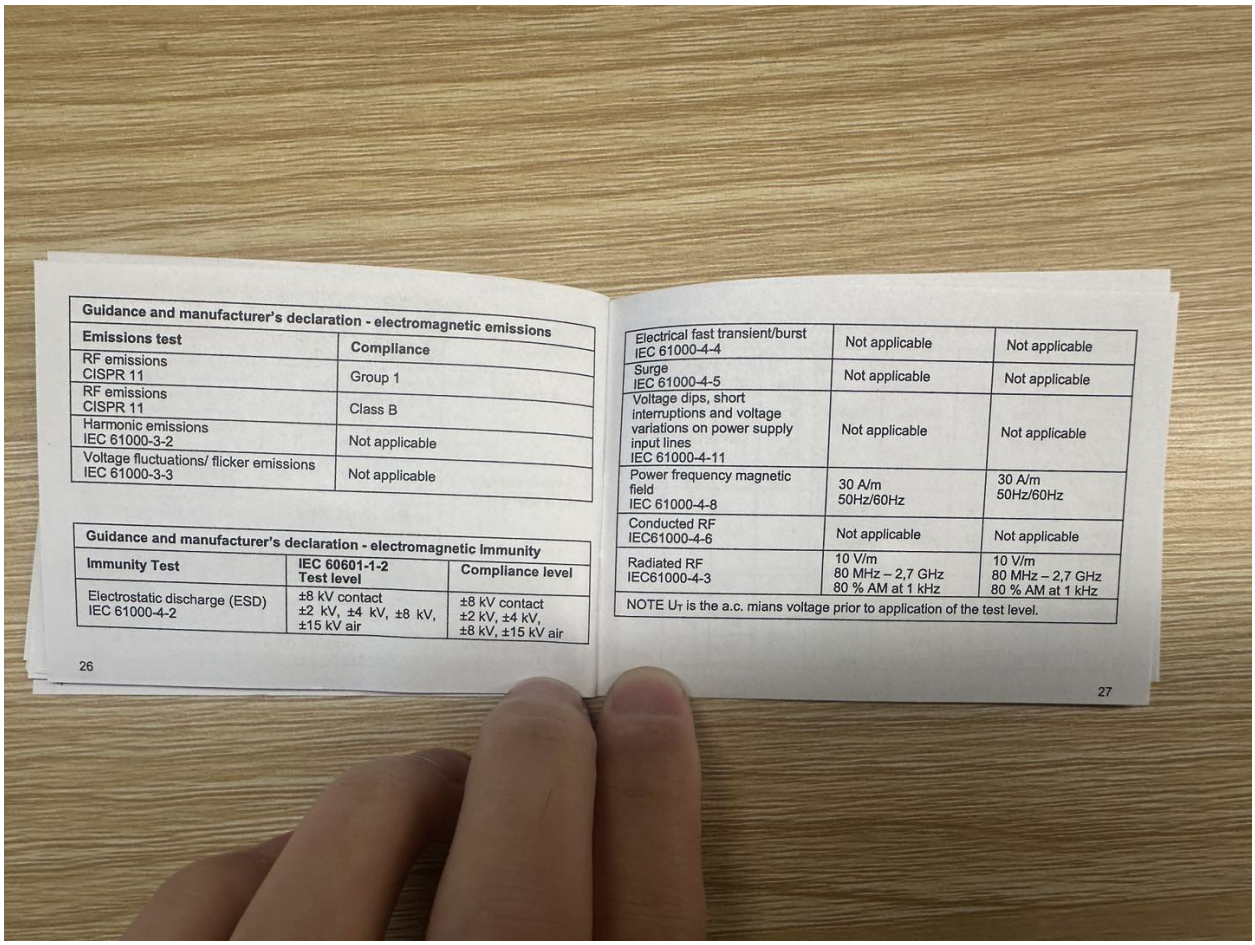
• When the measurement is abnormal, the following symbols will appear. Please use the correct method for measurement.

Error Indicators	Cause/Correction
Er U	The inflation can not reach 30 mmHg in 7 seconds.
Er H	The inflation reaches 295mmHg.
Er 1	The pulse rate is not detected correctly.
Er 2	Move, talk, or magnetic disturbance during a measurement.
Er 3	The test result is abnormal.
Er 23	SYS value reads lower than 45mmHg.
Er 24	SYS value reads hinger than 230mmHg.
Er 25	DIA value reads lower than 25mmHg.
Er 26	DIA value reads hinger than 160mmHg

TROUBLESHOOTING

Anomaly	Inspection Items	Countermeasures
There is no display after power on	Whether the battery power is insufficient	Replace the battery
	Whether positive and negative poles of the battery are installed reversely	Install in the positive and negative pole direction of battery
No pressurizing	Whether the air tube plug is inserted tightly	Insert the air tube plug firmly into the jack
	Whether the air tube is broken or leaked?	Purchase a new cuff
Unable to measure due to display error	Whether the arm is moved when pressurization	Keep your arm and body still static
	Whether you talk during measurement	Keep quiet while measuring the blood pressure
Air leakage of cuff	Whether the cuff is twined too loose	Please tighten the cuff
	Whether the air bag of cuff is ruptured	Replace with new cuff

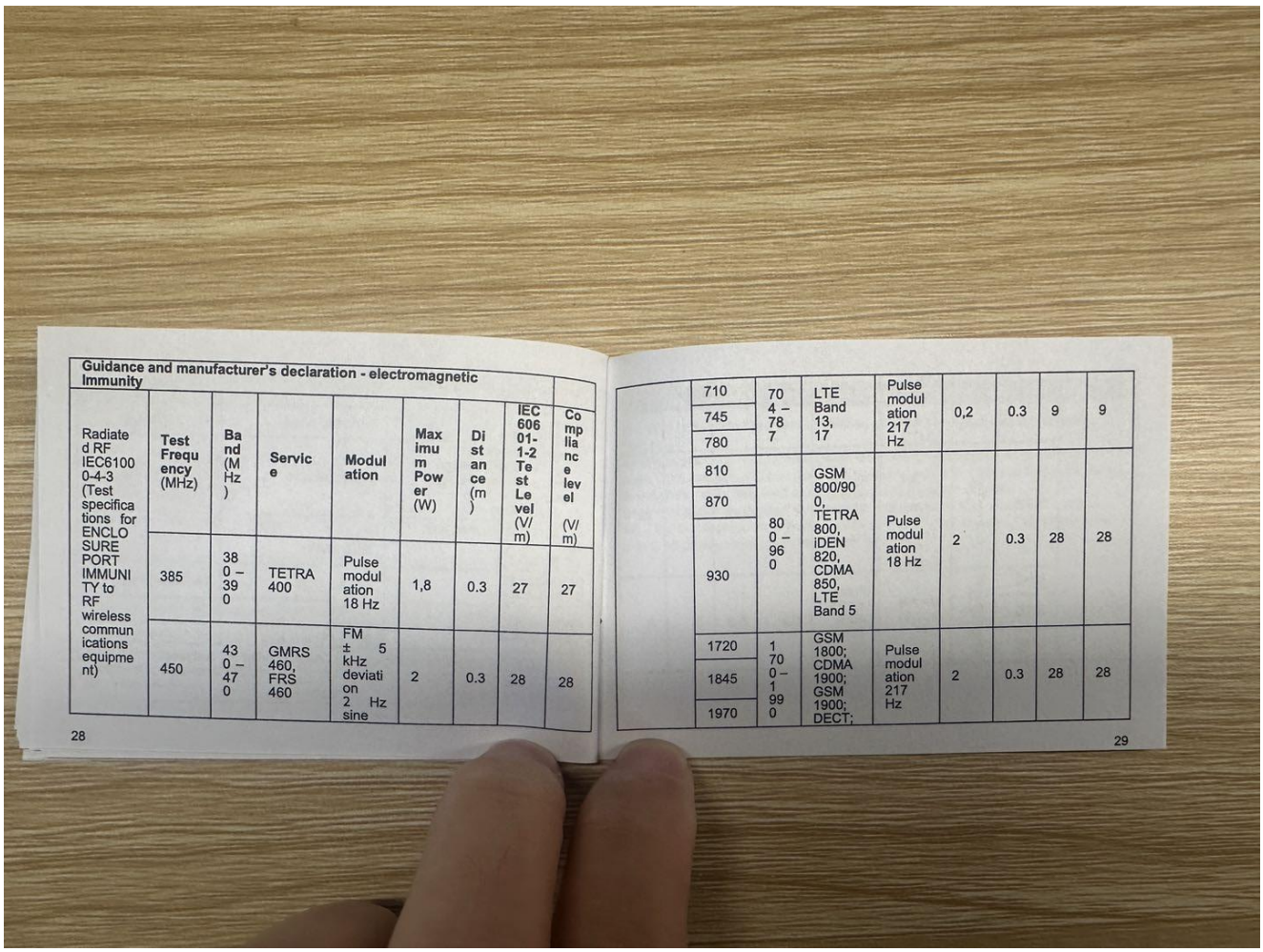
⚠ If the blood pressure still cannot be measured normally by the above methods, please contact the vendor. Please do not disassemble it by yourself!



Guidance and manufacturer's declaration - electromagnetic emissions	
Emissions test	**Compliance**
RF emissions CISPR 11	Group 1
RF emissions CISPR 11	Class B
Harmonic emissions IEC 61000-3-2	Not applicable
Voltage fluctuations/ flicker emissions IEC 61000-3-3	Not applicable

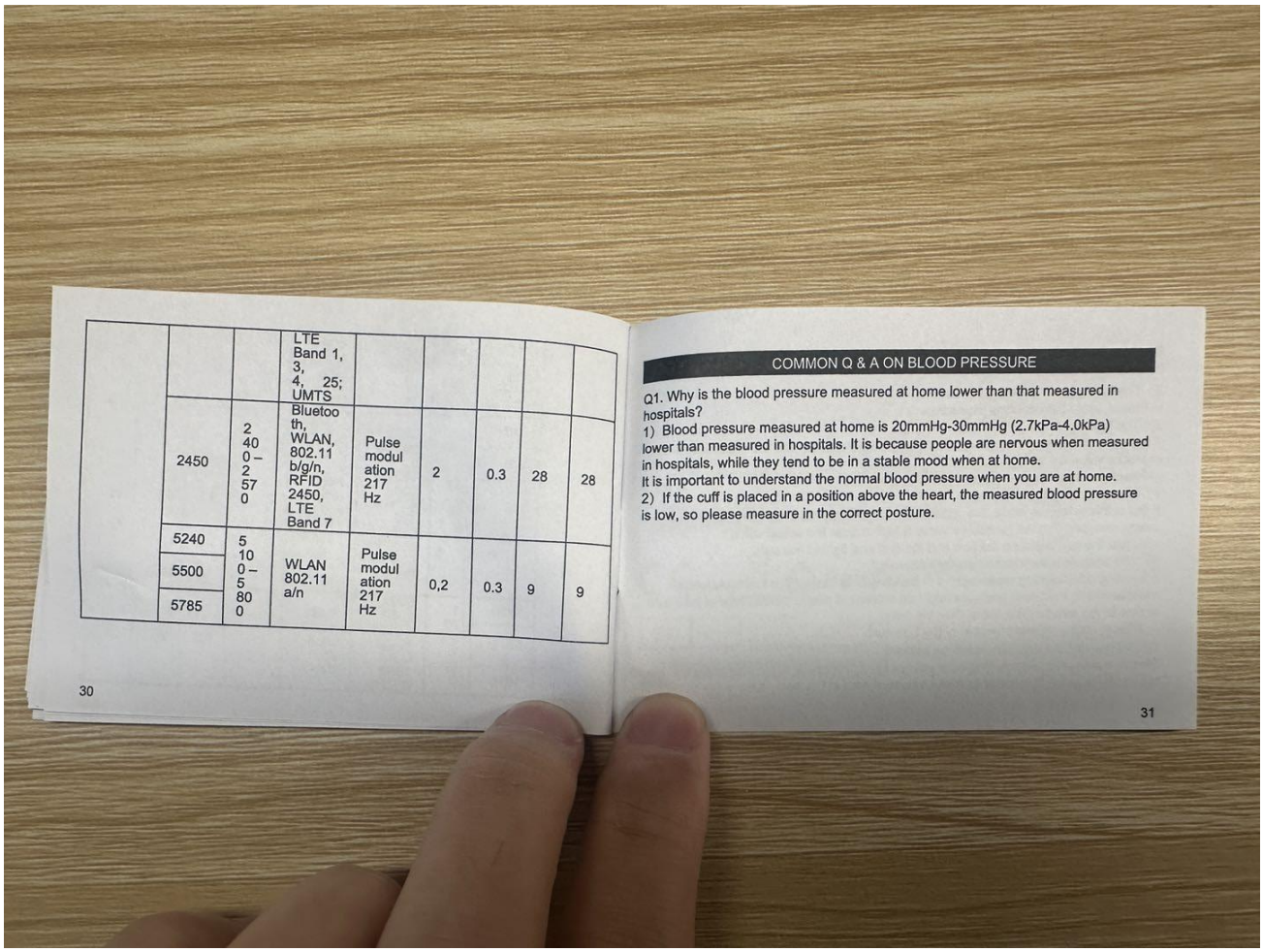
Guidance and manufacturer's declaration - electromagnetic Immunity		
Immunity Test	**IEC 60601-1-2 Test level**	**Compliance level**
Electrostatic discharge (ESD) IEC 61000-4-2	±8 kV contact ±2 kV, ±4 kV, ±8 kV, ±15 kV air	±8 kV contact ±2 kV, ±4 kV, ±8 kV, ±15 kV air

Electrical fast transient/burst IEC 61000-4-4	Not applicable	Not applicable
Surge IEC 61000-4-5	Not applicable	Not applicable
Voltage dips, short interruptions and voltage variations on power supply input lines IEC 61000-4-11	Not applicable	Not applicable
Power frequency magnetic field IEC 61000-4-8	30 A/m 50Hz/60Hz	30 A/m 50Hz/60Hz
Conducted RF IEC61000-4-6	Not applicable	Not applicable
Radiated RF IEC61000-4-3	10 V/m 80 MHz – 2,7 GHz 80 % AM at 1 kHz	10 V/m 80 MHz – 2,7 GHz 80 % AM at 1 kHz
NOTE U_T is the a.c. mians voltage prior to application of the test level.		



Guidance and manufacturer's declaration - electromagnetic immunity

Radiated RF IEC61000-4-3 (Test specifications for ENCLOSURE PORT IMMUNITY to RF wireless communications equipment)	**Test Frequency** (MHz)	**Band** (MHz)	**Service**	**Modulation**	**Maximum Power** (W)	**Distance** (m)	**IEC 60601-1-2 Test Level** (V/m)	**Compliance level** (V/m)
	385	380 – 390	TETRA 400	Pulse modulation 18 Hz	1,8	0.3	27	27
	450	430 – 470	GMRS 460, FRS 460	FM ± 5 kHz deviation 2 Hz sine	2	0.3	28	28
	710	704 – 787	LTE Band 13, 17	Pulse modulation 217 Hz	0,2	0.3	9	9
	745							
	780							
	810	800 – 960	GSM 800/900, TETRA 800, iDEN 820, CDMA 850, LTE Band 5	Pulse modulation 18 Hz	2	0.3	28	28
	870							
	930							
	1720	1700 – 1990	GSM 1800; CDMA 1900; GSM 1900; DECT;	Pulse modulation 217 Hz	2	0.3	28	28
	1845							
	1970							



			LTE Band 1, 3, 4, 25; UMTS					
	2450	2400 – 2570	Bluetooth, WLAN, 802.11 b/g/n, RFID 2450, LTE Band 7	Pulse modulation 217 Hz	2	0.3	28	28
	5240	5100 – 5800	WLAN 802.11 a/n	Pulse modulation 217 Hz	0,2	0.3	9	9
	5500							
	5785							

COMMON Q & A ON BLOOD PRESSURE

Q1. Why is the blood pressure measured at home lower than that measured in hospitals?

1) Blood pressure measured at home is 20mmHg-30mmHg (2.7kPa-4.0kPa) lower than measured in hospitals. It is because people are nervous when measured in hospitals, while they tend to be in a stable mood when at home.
It is important to understand the normal blood pressure when you are at home.

2) If the cuff is placed in a position above the heart, the measured blood pressure is low, so please measure in the correct posture.

Q2. Why is the blood pressure measured at home higher than that measured in hospitals?

1) The anti-hypertensive drugs lose the efficacy.
→Please accept the doctor's instruction.
2) The cuff is not twined correctly.
If the cuff position is wrong, no arterial signal will be caught, and the measured blood pressure value will be higher.
→Please confirm whether the cuff position is correct.
3) The cuff is rolled too loose.
If the cuff is rolled too loose, the compression force may fail to be transmitted to the artery, resulting that the measured value is higher than the actual value.
→Adjust the gap between the arm and the cuff and tighten the cuff.
4) The posture is incorrect when measurement.
If bending down, sitting cross-legged or bending over the sofa or low table when measurement, the blood pressure might be increased due to the abdominal pressure or due to the arm position lower than the heart.
→ Please make measurement in correct posture.

Q3. When will the measurement be better?

Please measure in the morning after you get up to urinate or when your body and mood are in stable state. It is better to make measurement at the same time every day.

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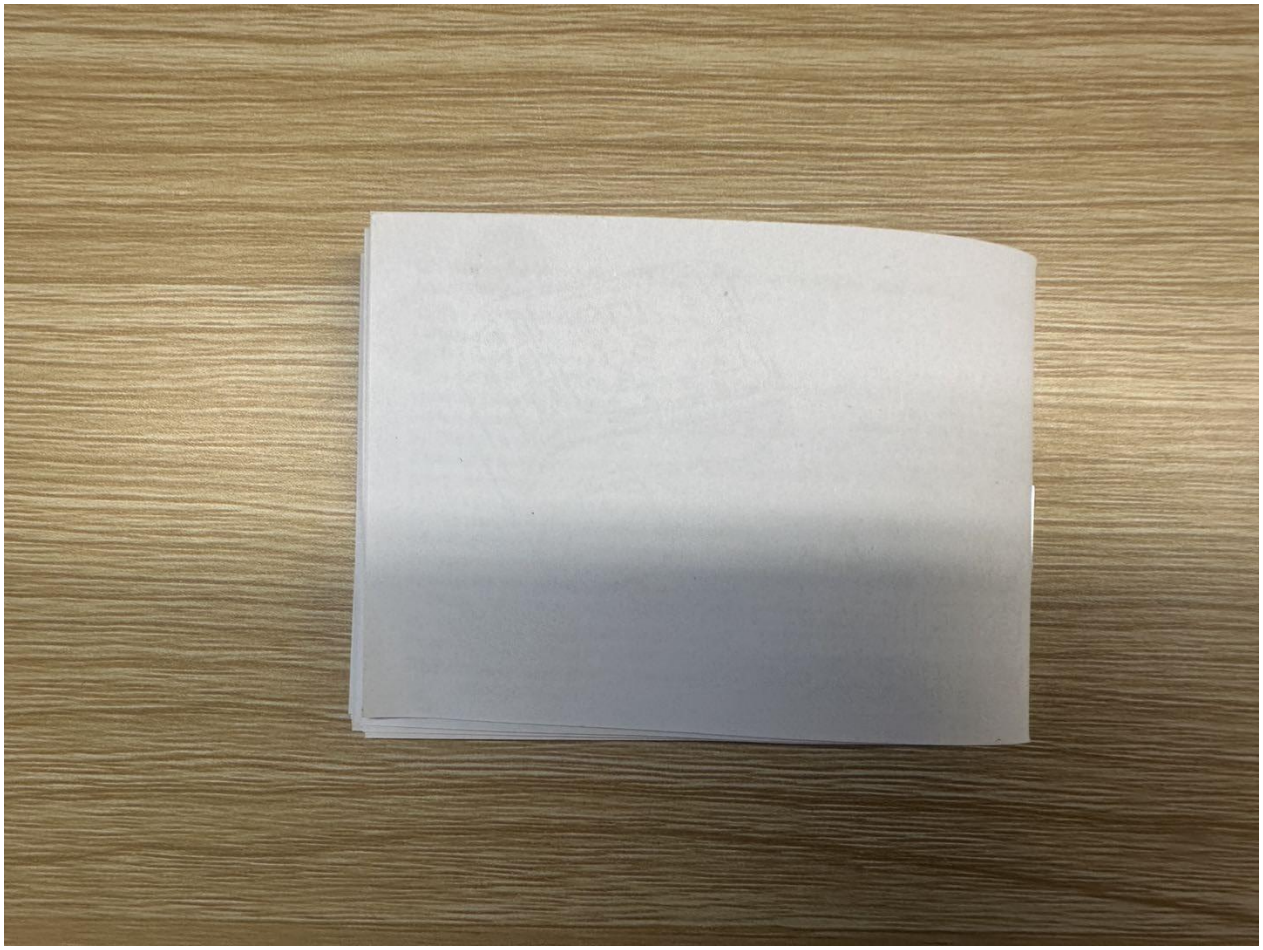
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